

Introduction

Transformers display a remarkable in-context learning (ICL) ability: given a prompt that specifies a task, a fixed-weight model can implement the task at inference time without updating its parameters, an ability that has been attributed to the forward pass implementing gradient descent or ridge regression on the prompt [3,16]. The same capability extends to sequential decision-making. Decision transformers and supervised-pretrained policies act by conditioning on a trajectory of past states and actions [1,2], and recent theory shows that a single transformer layer can provably implement reinforcement learning (RL) in-context: with linear attention the forward pass is exactly a semi-gradient SARSA or Q-learning update and yields provable policy improvement [4], whereas with softmax attention the forward pass implements a softmax-weighted non-parametric TD update for policy evaluation only [5]. Throughout this paper we use the following running example: an agent acts in an unknown tabular MDP and observes a growing trajectory of transitions (s, a, r, s') . In-context RL frames policy refinement as a single forward pass of a fixed-weight attention layer over this trajectory; the layer must turn accumulated experience into a better policy.

Despite this progress, three limitations remain. First, the constructive policy-improvement results all assume linear or ReLU attention [4,6,7], whose unnormalised weights may take any sign; these results do not transfer to the softmax-normalised attention used in practice, where the guarantees degenerate to evaluation or disappear. Second, for softmax attention the constructive theory stops at evaluation: existing analyses show that softmax layers estimate value functions in-context [5,12], and that softmax attention emulates first-order methods only in kernel spaces, where the sign of the update is supplied by an external label rather than derived from an endogenous temporal-difference (TD) error [8,9,10]. To our knowledge, the capability "a softmax attention layer performs a signed policy-improvement step" is unclaimed; bandit-style self-improvement under Fisher-weighted logit matching is confined to linear attention [11].

The difficulty is intrinsic. Policy improvement requires a signed TD update, $\delta = r + \gamma \max_{a'} Q(s', a') - Q(s, a)$, whose sign must survive the update. Softmax normalisation confines attention outputs to the convex hull of the input tokens — weights are non-negative and sum to one — which is structurally incompatible with signed updates [13]; moreover, softmax sharpness degrades with sequence length [14]. The hard constraint on any constructive softmax improvement mechanism is therefore to express a signed update through positive, normalised weights, without relying on an unnormalised bilinear term. We ask: can a single softmax attention layer implement Q-learning-style policy improvement in-context, and, if so, what is the precise boundary of what it can express?

A naive extension of softmax evaluation fails for three reasons. First, the convex-hull constraint: a softmax-weighted average of token values can move only toward the interior of the convex hull of the values, so it cannot subtract a negative contribution; the sign of δ is lost in the normalisation and the update deadlocks. Second, self-containment of the current value: the improvement residual requires a live value $Q(s, a) = \phi(s, a)^T w$, yet a softmax block cannot exactly re-derive this bilinear form — in our measurements it reconstructs it only partially, at a structural floor of $R^2 \approx 0.92$ — because normalisation discards the unnormalised $\Phi^T \Phi w$ term that linear attention can carry [15,16]. Third, on-policy bootstrap deadlock: bootstrapping against the current policy drives the expected TD error to zero by Bellman consistency, so the improvement signal vanishes unless a greedy (max) bootstrap breaks the deadlock.

We address these challenges with a score-scaled softmax improvement operator. Instead of feeding the TD error through a value-weighted readout, the layer multiplies the TD errors back into the attention weights, accumulating $\sum_t \delta_t \cdot \text{softmax}(\delta_t / \tau) \cdot \phi_t$; the sign of the update is recovered from the multiplied-back δ_t . As $\tau \rightarrow \infty$ this operator degenerates exactly to the semi-gradient Q-learning update $(1/n) \sum_t \delta_t \phi_t$ of [4]. A greedy max bootstrap breaks the on-policy deadlock, and the evaluation value is supplied externally or reconstructed up to the measured floor $R^2 \approx 0.92$. On random tabular MDPs, a single softmax layer with a linear readout achieves closed-loop returns of 0.42–0.45, matching the numpy semi-gradient Q-learning upper bound (0.428–0.461) and exceeding one-step greedy behaviour (0.376); placing the score-scaled operator inside the attention weights reaches the same level for $\tau \geq 5$.

We summarise our contributions as follows. (i) We give the first constructive demonstration that softmax attention can implement Q-learning-style policy improvement in-context: a single layer with a linear readout achieves closed-loop returns of 0.42–0.45 on random tabular MDPs, matching semi-gradient Q-learning and exceeding one-step greedy behaviour (Sections 4 and 5). (ii) We show that the improvement operator can be placed inside the attention weights via score-scaled softmax, $\delta \cdot \text{softmax}(\delta / \tau)$, which degenerates exactly to semi-gradient learning as $\tau \rightarrow \infty$ (Section 4). (iii) We characterise the structural boundary of softmax improvement: the current value $Q = \phi^T w$ cannot be self-contained — softmax normalisation discards the unnormalised bilinear term, leaving a reconstruction floor of $R^2 \approx 0.92$ — and we verify the operator's failure modes under sparse-reward grid worlds (Sections 5 and 6).

References

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